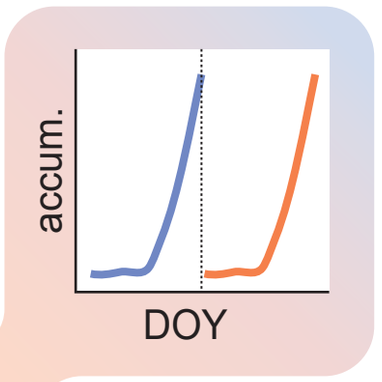
Unmeasured between dormancy start and budbreak



sequential



Dormancy

‘heat’ accumulation
parallel response

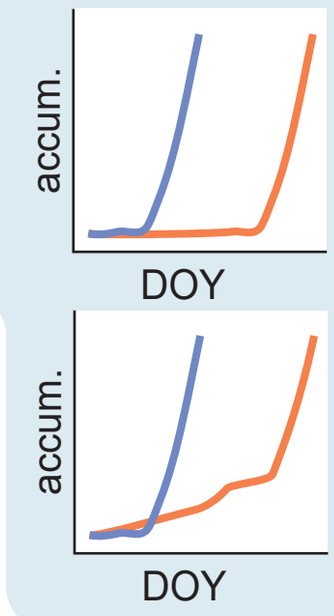
‘Forcing’
‘ecodormancy’ phase



Budbreak

‘Chilling’
‘endodormancy’ phase

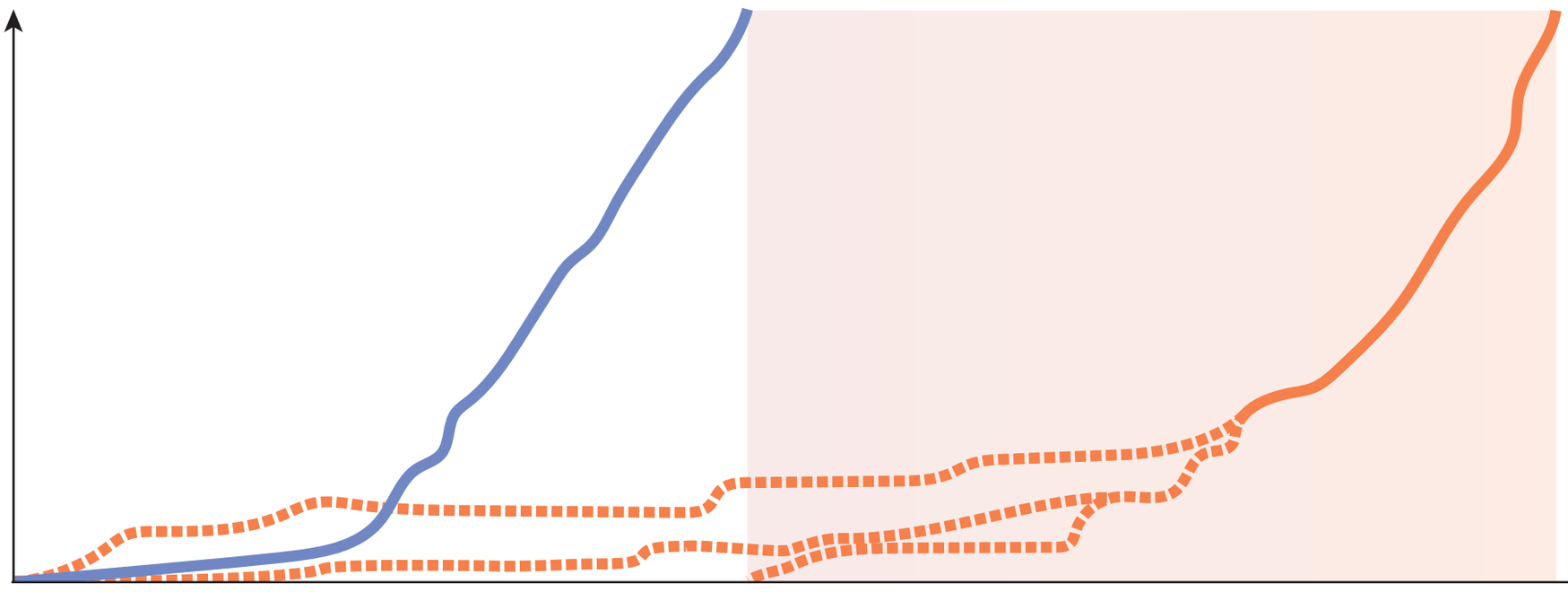
‘chill’ accumulation
parallel response



overlapping

Models

‘Chill’ and ‘heat’ accumulation



Callose deposition, hydrolysis

GH17 glucanase family

Beta-glucanase
Bpβ-glucanase (24-kDa)

Callose synthase/
glucan synthase-like
PmGSL3
PpGSL1,2,7,8

Gene expression

GA biosynthesis/ ABA catabolism

GA catabolism
PmGA2OX

ABA catabolism
PpABA8’OH

GA biosynthesis
PmGA3OX1
PmGA20OX8

Flowering time, master regulators

Dormancy regulation
PpDAM1

Floral development
GtFT2

Budbreak regulation
PtEBB1

Floral repression
AcFLC
MaMADS33 (FLC-like)

Dormancy onset

Endodormancy

Endodormancy release

Ecodormancy

Growth resumption

Seasonal Progression